# How AI Lowered the Barrier to Rigorous Science, and Why It Still Needs a Human Who Has Lived in the World

*An Inquisitive Amateur, an AI Pair Programmer, and JPL Space Race Heritage Audit*
*Raw NASA Telemetry on an Anomalous Cylinder in Gale Crater*



The object in context: a small cylinder in the Martian regolith.

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### It started with a picture

In August 2022, NASA's *Curiosity* rover took a photo on Mars. In the lower right corner was something strange. There appeared to be a cylinder poking out of the dirt. It's surrounded by pebbles and dirt, things that one would expect to find on Mars. The object was, at least to me, obviously anomalous.

People noticed. Some said it was debris from the rover. Some said it was just a strange rock. And of course some people immediately jumped to aliens. Even astrophysicist

Avi Loeb wrote that NASA should take a closer look.

I'd seen it discussed on podcasts and YouTube videos, but no one ever seemed to actually meaningfully dig into the data. No one seemed to be doing any actual work, just making a lot of noise. This applies both directions, from the scientific community and the more heterodox crowd.

I began this project wanting to determine the actual dimensions and whether or not it was mathematically cylindrical. This seemed relatively straightforward and in my mind would bring some clarity to the discussion.

However, seeing familiar shapes in chaos, and our desire to see what we want to see is a very real, uniquely human trait. I can watch a podcast, see a photo someone presents of a "pyramid" or a face, and just because of who I am I am inclined to believe it. This is likely why the scientific community is skeptical of such claims. So I had to constrain myself to the raw telemetry, the resulting data, and nothing else.

On that note, a disclaimer: I'm not an engineer and I'm not a mathematician. I've never taken a course in computer vision or planetary photogrammetry. However, I am inquisitive and stubborn, probably have too much time on my hands, and happen to have a fun family connection to this work.

**A family footnote on Mars**

This story begins with another story: one that makes what is to come more interesting. Not more credible, but certainly more entertaining. We begin in Decatur, Indiana, in 1931. My great-grandmother Flossie gave birth to a baby boy, Richard "Sam" Bogner, whose work would, a few decades later, make the work I have done today possible. Named after his father, my great-uncle Sam would eventually graduate from Ball State Teachers College in Muncie, now Ball State University, in 1959. His career as a chemical engineer began with Delco Remy, a major automotive electrical parts manufacturing division of General Motors located in Anderson, Indiana. In 1966, he joined the exodus of young men from this region, an area still known for producing top-tier engineering talent, who took their degrees to the front lines of the space race. Sam relocated his family to Pasadena, California, to work at NASA's Jet Propulsion Laboratory, or JPL.

During his time at JPL, he worked on evaluating and perfecting the high-energy silver-zinc battery packs that would power the Surveyor lunar landers. It wasn't just about making the batteries hold a charge; his team had to ensure these batteries could withstand the violent vibrations of a rocket launch, survive the deceleration of a lunar touchdown, and keep the spacecraft alive through freezing, two-week-long lunar nights where temperatures plummeted to hundreds of degrees below zero.

From the Moon, his focus shifted outward to the Red Planet. He became the project engineer and the definitive voice on battery flight performance for the Mariner program, specifically Mariners 6, 7, 9, and 10. I looked up the historic NASA technical reports from the Mariner Mars 1971 mission, and seeing his name right there on the cover was profoundly humbling. His work tracked how deep-space silver-zinc and nickel-cadmium cells degraded over months of interplanetary transit, managed the chemistry of separator systems to prevent short circuits, modeled the exact discharge curves required when Mariner 9 entered orbit and dipped into the dark shadow of Mars, and a bunch of other stuff I still don't fully understand. When Mariner 9 successfully outlasted a planet-wide dust storm to map 100% of the Martian surface, it did so powered by an architecture

he engineered.

As the space race began to transition into the commercial satellite boom of the late 1970s and 1980s, Uncle Sam brought his expertise to Hughes Space and Communications. At Hughes, he moved away from the volatile inner solar system and focused on longevity and weight reduction for Earth's orbital infrastructure. He pioneered research into lightweight, graphite-epoxy battery casings to replace heavy metallic housings, testing nickel-cadmium cells through simulated geosynchronous orbit-eclipse seasons to ensure communication satellites could survive for decades without failing.

This work ties directly into the next portion of his career, and takes him back to his beginnings in the automotive industry. In 1987, General Motors partnered with AeroVironment and Hughes to build the SunRaycer, a revolutionary solar-powered vehicle designed to contest the inaugural World Solar Challenge across the Australian Outback. On the SunRaycer project, he functioned as a primary battery design engineer and lead technical researcher representing Hughes Aircraft. He is officially credited in the Society of Automotive Engineers (SAE) archives as one of the definitive engineers of the vehicle's power architecture, co-authoring the official technical documentation titled: *"GM Sunraycer Case History Lecture 3-4: Design and Performance of the GM Sunraycer Battery"* alongside his colleagues N. Chorneau and D. Pickett. In this capacity, he helped engineer a custom, ultra-lightweight silver-zinc secondary battery pack. This pack had to flawlessly store transient energy harvested from the car's cutting-edge gallium-arsenide solar array, delivering bursts of raw power when navigating hills or cloudy stretches. And he unequivocally delivered. The engineering was so pristine that the GM SunRaycer didn't just win the 1,800-mile race across the continent. It crossed the finish line more than two full days ahead of the nearest competitor. The car currently sits in the Smithsonian in Washington, D.C.

I would love to say that I grew up listening to my great-uncle Sam tell stories about NASA, rocket ships, or the space race. Unfortunately, I don't recall ever meeting him. My grandpa Alan mentioned his older brother from time to time, but as a kid, I don't think I ever paid much attention. Then one day, less than a year ago, my dad casually mentioned his work on the SunRaycer. I was beyond intrigued. That was when I discovered all of this for the first time. So, while I was never regaled with stories of his work in space—which, even as a child, I would have found profoundly interesting—I hope the work that follows, enabled by his early research at JPL, might continue, in a much smaller way, my family's legacy on the Red Planet.

I didn't think I could ever do anything as interesting. It turns out that today, I could do a surprising amount of it. NASA publishes raw images and telemetry from its rovers, including pointing direction, focus distance, and solar angles. And modern AI, when rigorously guided by human oversight, is much better at math than I will ever be.

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### What I actually did: Dissecting the data

Before you can ask what something is, you have to find out what it actually measures. Here is what happened when we dug into NASA's raw files:

- **We busted the "20-centimeter" rumor.** Online commentators claimed the object was 20 centimeters long by assuming it was sitting on a distant cliff 45 meters

away. But NASA's telemetry logs showed the camera's focus motor count was set to 2705, meaning the lens was focused at exactly **3.96 meters** from the rover. The object isn't on a distant cliff; it's sitting right in front of Curiosity. That dropped its true dimensions to roughly **3.5 centimeters long and 11 millimeters across** or about the diameter of a AA battery.

- **We tested the shadow with solar math.** The mission records showed the Martian Sun was sitting at an elevation of 69.8° (nearly straight overhead at noon). At that angle, an object's shadow on the ground must be compressed to just 37% of its physical height. When we measured the shadow fringe beneath the cylinder ($9.8 \pm 0.7$ pixels) and compared it to what the Sun's position dictated, both optical scales—the normal transverse camera scale (0.293 mm/px, yielding 7.81 mm relief and a 10.76 mm full diameter) and the inclined ground slope scale (0.317 mm/px, yielding 8.46 mm relief and an 11.66 mm diameter)—bracket the physical diameter to $11.2 \pm 0.5$ mm ($< 8.3\%$ spread). That proved beyond doubt that this is a real, rigid, three-dimensional physical object casting a genuine shadow on the soil, not a digital smudge, a compression defect, or an optical illusion.
- **We pulled the boundaries apart with edge filters.** To keep my eyes from tricking me, I had the AI write code applying spatial "Sobel gradient filters." This math measures how abruptly brightness changes across pixels. It proved that the dark circle on the end wasn't a shadow thrown on the ground, it was the cylinder's own flat, self-shadowed face, completely separate from the neighboring rock and the dirt beneath it.

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### Building the tool and finding the back edge

Ten years ago, I couldn't have done this analysis because the tools were out of reach. Reading NASA's data format, working out camera geometry, writing code for edge detection, fitting ellipses, running the error checks… each of those is a specialty.

AI collapsed that barrier. It wrote the Python code that reads the data format and calculates the center of the image. It explained what a "Sobel filter" is. It did the arithmetic I can't do and the geometry I never learned. When I asked "what does that mean?", it explained, as many times as I needed.

**The barrier to rigorous analysis is lower than it has ever been.**

But here's the part people skip. **The AI got things wrong. Repeatedly.** It was honestly exhausting and at some points infuriating. The things that it couldn't seem to do, even with its ability to do calculus and trigonometry in seconds, were to me, dead simple.

It kept making the same kinds of mistakes:

- **It mixed up two different shadows.** The cylinder's end face is dark because it's turned away from the Sun. That's *its own* shading. Separately, the cylinder throws a shadow on the ground. The software saw "dark" and lumped them together, then built measurements on a shape that didn't exist. I could see, just by looking, that those were two different things.

- **It tried to fit a perfect circle to something that wasn't one.** When you look at a circle from an angle, it becomes an ellipse. The AI kept forcing a circle onto a tilted view. It's the kind of mistake that most intelligent people wouldn't make, mostly because they have spent a lifetime of looking at shadows and shapes in the world.
- **It swallowed a neighboring rock.** Next to the cylinder was a lump that wasn't part of it. The software kept including it in the outline until I told it to leave that part out.

I finally had the AI build a custom web-based photogrammetry tool that allowed me to overlay a physically accurate 3D cylinder directly onto the NASA photo and manipulate it in real-time. It was as simple as using a mouse wheel to zoom and dragging the mouse to adjust rotation, perspective tilt, and length.




Interactive 3D photogrammetric overlay tool showing the cylinder fitted to the Sol 3556 Mastcam-100 imagery.

That's when the breakthrough happened:

By dragging the length slider and locking the perspective tilt to match Curiosity's 22° downward camera angle, **the back edge of the cylinder lined up with the rock highlight.**

People assumed the cylinder was either a broken stub or a continuous rod plunging indefinitely into the Martian dirt. But the 3D model revealed that the rear cap terminates right where the highlight ends and meets the regolith. It is an exposed, self-contained,

discrete cylinder: **110 pixels by 34 pixels**, giving an exact length-to-diameter aspect ratio of $3.24 : 1$ ($\approx 35$ mm $\times$ 11 mm).

Notice what I contributed. It wasn't equations. It was **judgment**. I knew what a shadow looks like. I knew how a cylinder behaves. I knew when something looked wrong. The AI had the mathematical ability. I had the *common sense* to tell it when it was pointed in the wrong direction.

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### Embodied experience still matters

I think this is the most important thing I learned, and I'd like you to take it away from this article.

An AI has read about shadows. I have *stood* in them. It knows the word "perspective." I've driven through a tunnel and seen the end getting closer. It can describe a coin on a table. I've flipped one and watched the circle become an ellipse.

That lived, physical understanding of how things look and behave is not a small thing. It's what let me notice, quickly and without any formal training, when the analysis had drifted away from reality.

AI can be confidently, fluently wrong. It can produce a clean chart and a convincing explanation of a result that's broken.

The people who are going to excel moving forward won't be the ones who trust AI the most or the least. They'll be the ones who combine **powerful tools with real-world wisdom** and stay in charge of the process.

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### Why this actually matters

Why does measuring a tiny 3.5-centimeter object on Mars matter?

1. **It replaces wild speculation with checkable facts.** In an online landscape divided between wild claims ("it's proof of an alien probe!") and lazy dismissal ("it's just a rock, stop looking"), numbers provide sanity. Anyone with a computer and curiosity can download NASA's Sol 3556 dataset, run the equations, and verify the dimensions for themselves. That is how science is supposed to work.
2. **It shows that curiosity doesn't need credentials to be rigorous.** You don't need a PhD or a corporate lab grant to audit space agency data. The data belongs to the public. If you are willing to learn, ask hard questions, and refuse to accept comfortable assumptions, you can do genuine planetary forensics from your living room.
3. **It demonstrates the future of human-AI collaboration.** AI isn't an oracle that hands down absolute truth, nor is it a useless novelty. It's an exoskeleton for human curiosity. When guided by someone who refuses to let it hallucinate, it allows one person to do the work that once required an entire research team.

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### You can do this too

NASA's rover images are public. So are images from the Hubble and James Webb telescopes, satellite archives, weather data, seismic logs, and thousands of other datasets. For most of history, serious investigation required a university, a lab, and a funding source. Now it requires **curiosity, patience, and a willingness to be corrected.**

If you're thinking about trying something like this, here's what I'd suggest:

1. **Start with a question you can actually answer.** Not "is this alien?" but "how big is this, really?" Small, measurable questions lead somewhere.
2. **Use the original data, not screenshots.** Compressed social media images throw away exactly the information you need.
3. **Let AI do the heavy lifting, but never the thinking.** Ask it to explain everything. Ask it to show its work. Ask "how would I know if this were wrong?"
4. **Trust your eyes and your instincts.** If a result looks off, make the software justify itself.
5. **Be honest about what you don't know.** The most credible sentence in my paper is the one that says that I don't know what the object is.
6. **Share your work openly.** Let other people check it, argue with it, and break it. That's how science works.

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### What happens next

I still don't know what that object is. But now I know that it's actually cylindrical. And I know that it shouldn't be there.

In the technical paper, I evaluated the three competing hypotheses:

#### 1. Terrestrial rover or landing debris

The most intuitive assumption is that Curiosity dropped a bolt, spacer, or piece of hardware, or that debris drifted from the 2012 Entry, Descent, and Landing (EDL) event. But this runs into severe physical problems:

- **The distance gap:** Curiosity's landing site (Bradbury Landing), where the sky crane and heat shield crashed, is over 28 kilometers away along the rover's winding drive path, and roughly 13 to 15 kilometers (8 to 9 miles) in a direct straight line through the air. In Mars's thin atmosphere (with less than 1% of Earth's air density), wind simply cannot loft and carry a solid structural cylinder across 13 kilometers of rugged mountainous terrain.

- **The sediment clock (80 to 200 years):** The cylinder isn't resting loosely on top of fresh dust. Its lower third ($\approx$ 2 mm of vertical relief) is embedded in the compacted soil matrix along a straight ground interface. In Gale Crater, background atmospheric dust settles at only about 10 to 25 $\mu$m per Earth year. Accumulating 2 millimeters of natural sediment fallout to bury that lower third requires roughly **80 to 200 years**. Curiosity had only been on Mars for 10 years at the time of the photo. Because Gale Crater has hosted no other mission in history, this passive dust clock rules out terrestrial spacecraft debris.
- **No mechanical trauma:** The cylinder shows no sheared mounting brackets, severed cabling, or buckling from a high-velocity ballistic impact ($R^2 = 0.9985$ straightness).
- **Tested against surrounding rocks:** When we verified an empirical control sample of twenty other rocks across the same frame (extracting each rock directly from the raw pixel detector), natural basalt and sandstone clasts averaged an aspect ratio of 1.23 and a generator straightness fit of only 0.10, with none scoring above 0.271. The cylinder's 0.9985 straightness is an extreme ten-sigma statistical outlier ($+10.70\sigma$, Cantelli bound $p < 0.009$).

**2. A wind-carved rock (ventifact) or mineral concretion**

The standard geological dismissal is that it is just a rock shaped by natural forces. But the math doesn't fit the geology:

- **What is a ventifact?** A ventifact (from the Latin for "wind-made") is a rock that has been sculpted, faceted, or polished by wind-blown sand over thousands or millions of years. Think of natural sandblasting. Because Martian wind blows primarily from dominant directions, saltating sand grains strike exposed surfaces, carving sharp ridges (keels), flat slanted faces (facets), and scooped grooves. You see them on Earth in places like Antarctica or the Mojave Desert.
- **Wind is not a lathe:** While natural sandblasting easily produces smooth stones or rounded pebbles (several natural rocks in the surrounding field score over 90% in projected roundness, meaning circularity alone is not an anomaly), it **does not act like a lathe**. The true anomaly is the non-parametric conjunction of three features that never appear together in natural geology:
  1. Two strictly straight, mutually parallel sides ($R^2 = 0.9985$ straightness, an extreme ten-sigma anomaly of $+10.70\sigma$, Cantelli bound $p < 0.009$, compared to the flattest natural rock in the scene at $R^2 = 0.2708$).
  2. A crisp, flat planar end cap terminating perpendicular to the body, whose 0.9335 apparent circularity perfectly matches the 22° camera perspective.
  3. An elongated 3.24 : 1 length-to-width aspect ratio (over twelve standard deviations, $+12.06\sigma$, Cantelli bound $p < 0.007$, away from the roughly equal dimensions of surrounding gravel).
- **Concretions grow into spheres, not cylinders:** Mineral nodules in Gale Crater (formed by chemical precipitation in groundwater) grow outward equally in all directions, creating spheres (like the famous Martian "blueberries") or lumpy clusters, not a clean elongated cylinder with an invariant cross-section.

### 3. An artifact of unknown provenance

Because the cylinder displays manufactured tolerances, yet cannot be easily explained by drifting landing debris or natural geology, scientific rigor demands keeping a third possibility on the table: that it could be an artificial artifact of unconfirmed origin.

When I say "something stranger," I mean something that most mainstream planetary science is hesitant to say out loud, but that every curious person wonders about in the quiet of their thoughts: could Mars have once hosted an ancient technological presence? If I am being completely candid about what drives my imagination, the spark of curiosity that kept me up late staring at pixel grids, I sincerely hope that this cylinder might turn out to be evidence of that.

Theoretical plasma physicist Dr. John Brandenburg pointed to isotopic Xenon-129 anomalies on Mars. Penn State astrophysicist Dr. Jason Wright published papers detailing how durable non-biological artifacts (technosignatures) could survive buried in planetary soil for billions of years. When I look at this clean cylinder with its flat circular end face, I think about those models.

**Do I know that's what it is? Absolutely not.**

And that honesty is what keeps this work credible. The math does not prove an ancient artifact; it only proves that the object is small, physically real, precisely shaped ($3.5\,\mathrm{cm} \times 1.1\,\mathrm{cm}$), and unaccounted for by conventional optical glitches or known rover shedding.

The honest answer is that the analysis narrows the possibilities, and a close-up look from Curiosity's contact instruments, including the MAHLI hand lens (resolving microscopic machining marks versus mineral crystal cleavage) and the ChemCam laser spectrometer (detecting alloy metals versus basaltic silicates), would do far more than any amount of image processing from Earth. That's what Avi Loeb asked for, and I agree with him.

The point isn't to jump to conclusions. The point is that *we are allowed to look, and we are allowed to wonder.*

Whatever the answer turns out to be, something has already changed. A person with no engineering degree and no math training was able to sit down with public data and produce a careful, checkable analysis.

The world is full of data. The tools are widely available. What it still needs is people willing to look carefully, ask honest questions, and use their own eyes and judgment to steer.

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*The accompanying technical paper,*
*"Morphological and Photometric Analysis of an Unidentified Cylindrical Feature in Curiosity Sol 3556 Mastcam Imagery," contains the complete methods, equations, and figures for anyone who wants to check the work, or improve on it.*